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Energy management and profit-sharing mechanisms for prosumer coalitions

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Overview

01 Introduction

02 Previous Work

03 Optimization Model

04 Profit Allocation

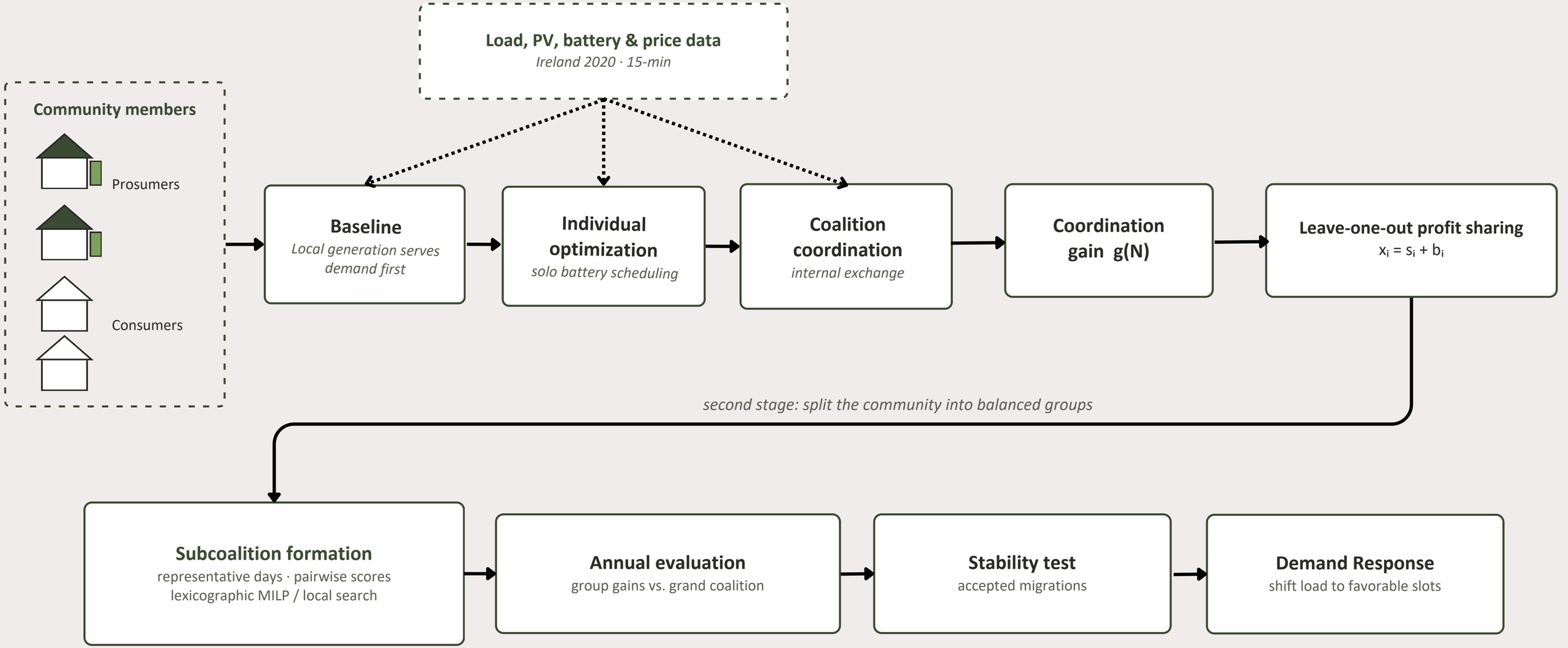
05 Subcoalition Formation

06 Subcoalition Stability under Profit Sharing

07 Demand Response

08 Conclusion

The Thesis in One Picture



Measure the value of cooperation from real operation → share it fairly → keep it in small, stable groups.

Introduction

Motivation

- Households are becoming active prosumers through PV generation, batteries, and smart meters.
  (Parag & Sovacool, 2016)
- Renewable energy communities coordinate local generation, storage, and consumption.
  (Barabino et al., 2023)
- Cooperation creates value by exploiting heterogeneous and complementary user profiles.
  (Tushar et al., 2020)
- Joint operation can reduce costly grid imports and improve local renewable-energy use.
  (Gu et al., 2022)

Problem Statement

- Model how users operate alone and how they cooperate inside an energy community.
- Measure the extra savings created by coordinated battery scheduling and internal energy exchange.
- Share these savings fairly and test whether smaller subcoalitions can remain efficient and stable.

Previous Work

Research strands

- Energy communities & local P2P sharing
  (Parag & Sovacool, 2016 · Tushar et al., 2021)
- Cooperative-game profit allocation
  (Shapley, 1953 · Schmeidler, 1969)
- Representative-period reduction
  (Kotzur et al., 2018 · Teichgraeber & Brandt, 2022)
- Coalition / group formation
  (Tushar et al., 2020 · Zhang et al., 2024)

→ This thesis connects the four strands in one operational framework.

The gap

- The strands are mostly studied as separate problems.
- Coalition values are often assumed abstractly — not derived from operational battery scheduling.
- Exact allocation and exhaustive partition evaluation do not scale. *(Li et al., 2020)*

Goals & Contributions

01 **Optimization framework** for baseline, individual, grand-coalition, and subcoalition operation.

02 **Profit allocation rule** based on marginal contributions, compared with cooperative game-theoritic rules.

03 **Scalable subcoalition formation** using pairwise scores and local search, with annual performance and stability evaluation.

04 **Demand Response** Targeted load shifting as an added flexibility source — quantifying its extra value on top of battery scheduling

Optimization Model

Baseline

- Local generation serves demand first
- Deficit imported from grid
- Surplus exported to grid
- Reference case, not an optimization
- Produces C_i^{base}

Solo scheduling

- Each household schedules its own battery
- No internal trading
- Objective: minimum individual cost
- Produces C_i^{solo}

Grand coalition

- All users optimize jointly
- Internal transfers q_i,j,t allowed
- Objective: minimum total cost
- Produces C_i^{coal}

MILP Formulation

Each optimization problem is implemented as a **Mixed-Integer Linear Program (MILP).**

Decision variables

- Battery charging / discharging and state of charge
- Grid import / export
- **Internal transfers q between coalition members — coalition only**

Compact objective

min

grid import cost − export revenue
+ battery degradation cost
+ internal-transfer regularization

$$\min \sum_t \left[\pi_t^{buy} g_t^{imp} \Delta t - \pi_t^{sell} g_t^{exp} \Delta t + \tfrac{\lambda}{2}(p_t^{ch} + p_t^{dis})\Delta t\right] + \varepsilon \sum_{i,j,t} q_{i,j,t}$$

Subject to

power balance · **battery dynamics** · **power limits** · **SoC bounds** · **terminal SoC constraint**

POWER BALANCE (+ TRANSFERS)

$$g_{i,t}^{imp} + p_{i,t}^{dis} + g_{i,t}^{PV} + \sum_j q_{j,i,t} = d_{i,t} + p_{i,t}^{ch} + g_{i,t}^{exp} + \sum_j q_{i,j,t}$$

BATTERY DYNAMICS · TERMINAL SOC

$$E_{t+1} = E_t + \eta\, p_t^{ch} \Delta t - \tfrac{1}{\eta}\, p_t^{dis} \Delta t, \quad E_T = E_0$$

POWER LIMITS · BINARY MODE

$$p_t^{ch} \le \bar{P} z_t, \quad p_t^{dis} \le \bar{P}(1 - z_t), \quad z_t \in \{0, 1\}$$

Binary mode variables prevent simultaneous battery charging and discharging · green terms appear only in the coalition problem.

Annual Evaluation



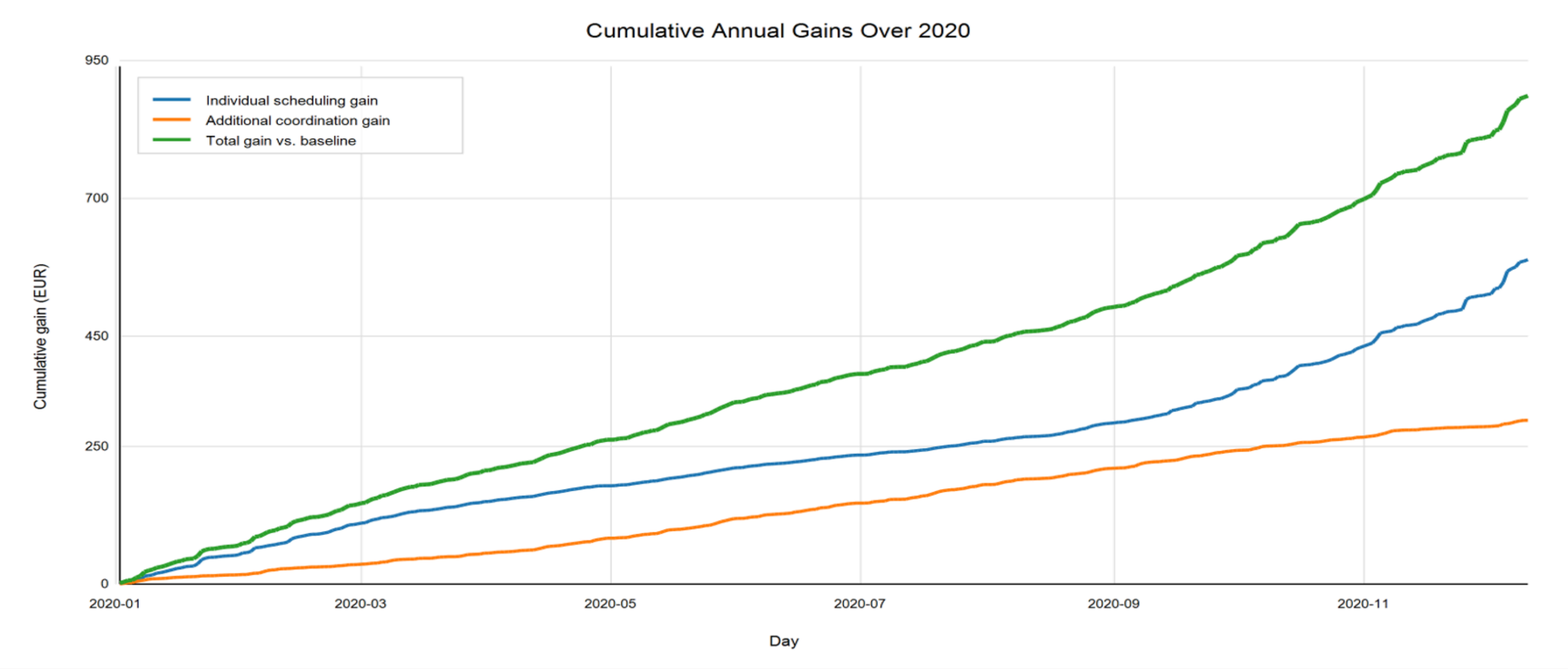
8.11%
solo scheduling gain vs. baseline

4.46%
coordination gain vs. solo scheduling

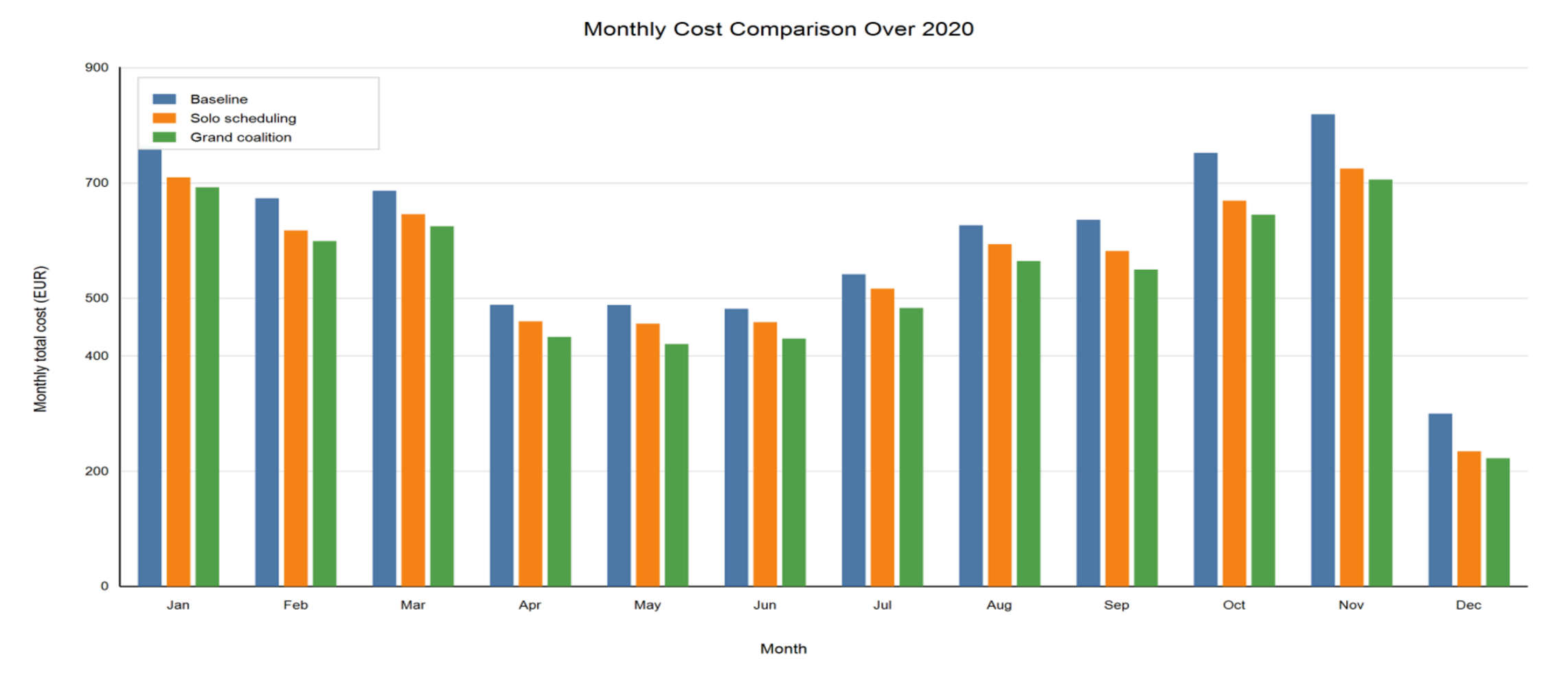
12.21%
total gain vs. baseline

Result: the grand coalition gives the lowest annual cost because it combines battery flexibility with internal energy sharing.

Annual Evaluation



Annual Evaluation



Cooperative Game & Profit Allocation

- Savings game: $v(S) = C^{base}(S) - C^{coal}(S)$, with stand-alone value $s_i = v(\{i\})$.
- Coordination-surplus game: $g(S) = \sum_{i \in S} C_i^{solo} - C^{coal}(S)$, isolating only the additional value of cooperation.
- Two value sources should be separated: individual battery benefit and coordination benefit.

Design principle

1. Each user keeps its guaranteed stand-alone benefit s_i.

2. Only the coordination surplus g(N) is allocated, proportionally to each user's contribution.

Proposed Rule: Leave-One-Out Contributions

01 $m_i = g(N) - g(N \setminus \{i\})$
Loss of coordination surplus when user i is removed from the grand coalition.

02 $w_i := m_i$
Non-negative contribution weight.

03 $b_i = g(N) \cdot w_i / \Sigma_j w_j$
Proportional allocation of the coordination surplus.

04 $x_i = s_i + b_i$
Final payoff: stand-alone benefit plus coordination bonus.

Computational cost: $2|N| + 1$ optimization problems, instead of $2^n - 1$ coalition evaluations for the exact Shapley value.

Properties of the Proposed Rule

Satisfied / by construction

✓ Efficiency / Budget Balance
$\sum_i b_i = g(N)$

✓ Individual rationality
$x_i \geq s_i$, since $b_i \geq 0$

✓ Non-negative cooperation bonus
$b_i \geq 0$ for every user

✓ Dummy-like behavior
$m_i = 0 \Rightarrow b_i = 0$

✓ Weak symmetry
$s_i = s_j$ and $w_i = w_j \Rightarrow x_i = x_j$

✓ Contribution awareness
$b_i \propto w_i$

✓ Contribution monotonicity
higher $w_i \Rightarrow$ higher b_i

Not guaranteed / by design

– Core stability
blocking subgroup deviations may still exist

– Additivity
not designed as an additive rule

– Population monotonicity
adding users can change weights

– Full Shapley symmetry/dummy/marginality
only leave-one-out terms used

Comparison with Classical Allocation Rules

Criterion	Proposed rule	Shapley value	Nucleolus	Equal split
Efficiency / budget balance	Yes	Yes	Yes	Yes
Individual rationality	Yes	Not guaranteed	Yes, if core ≠ ∅	Yes
Stand-alone value	Explicitly kept first	Implicit	Implicit	Not isolated
Contribution awareness	Leave-one-out g(N)	All marginal terms	Indirect, via excesses	No
Symmetry	Weak: same s_i, w_i imply equal x_i	Full Shapley symmetry	Satisfied	Trivial
Dummy / zero contribution	Dummy-like	Yes	No simple interpretation	No
Additivity	No	Yes	No	Yes
Core stability	Not guaranteed	Not guaranteed	Yes, if core ≠ ∅	Not guaranteed
Population monotonicity	Not guaranteed	Not guaranteed	Not guaranteed	Not guaranteed
Computational cost	Low: 1 + 2|N|	High: 2^|N|	High: sequential LPs	Trivial
Interpretability	High	Rich but complex	Less intuitive	High, ignores contribution

● satisfied ● weak / conditional ● not satisfied

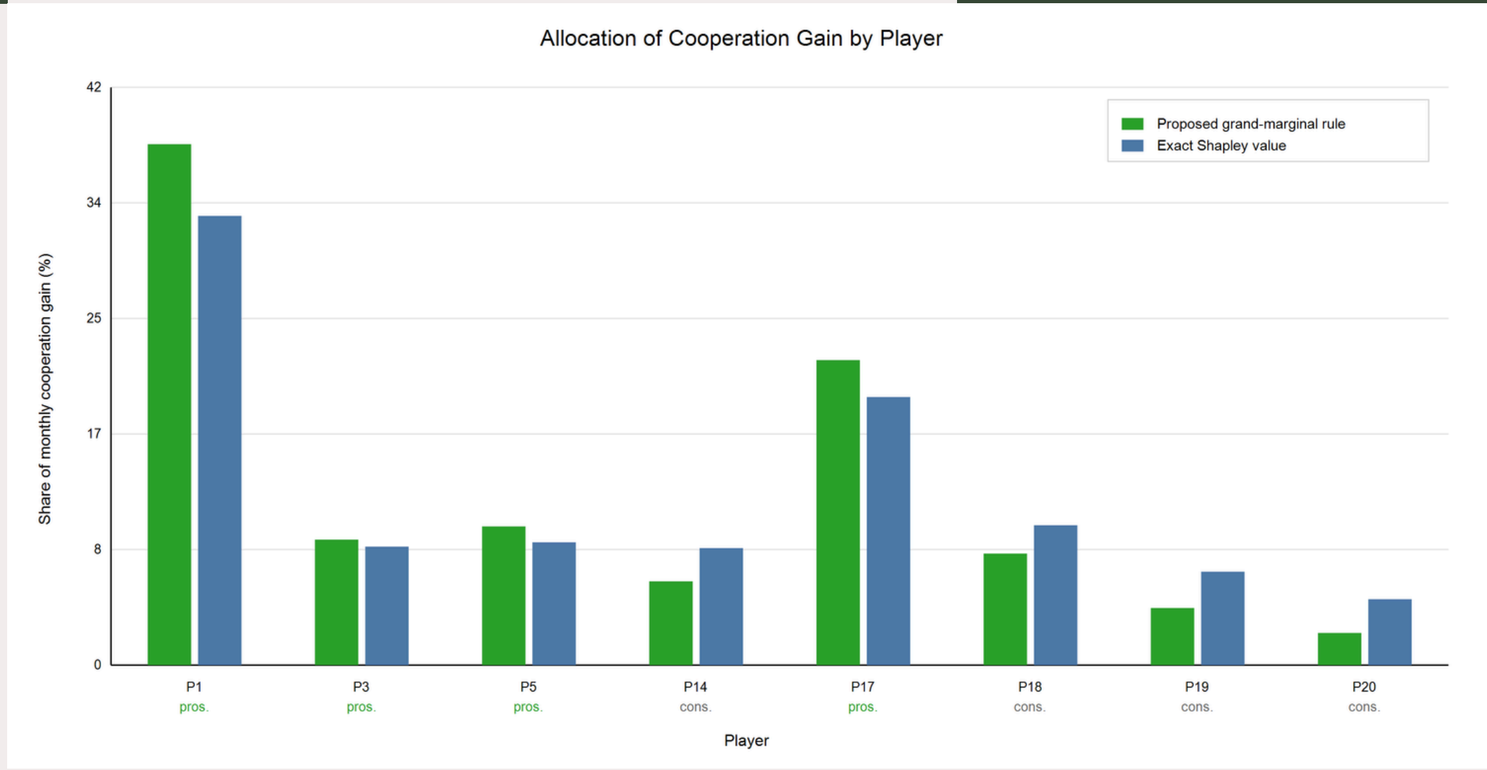
Takeaway: *despite its simplicity and low cost, the proposed rule satisfies most properties — at a fraction of the computation.*

Allocation test: proposed rule vs. Shapley

8-player test coalition

- Monthly cooperation gain is allocated with two rules.
- The proposed rule uses leave-one-out contribution weights.
- Exact Shapley value is used as a benchmark.

Result: the allocations are close, while the proposed rule gives slightly larger shares to high-contributing prosumers.

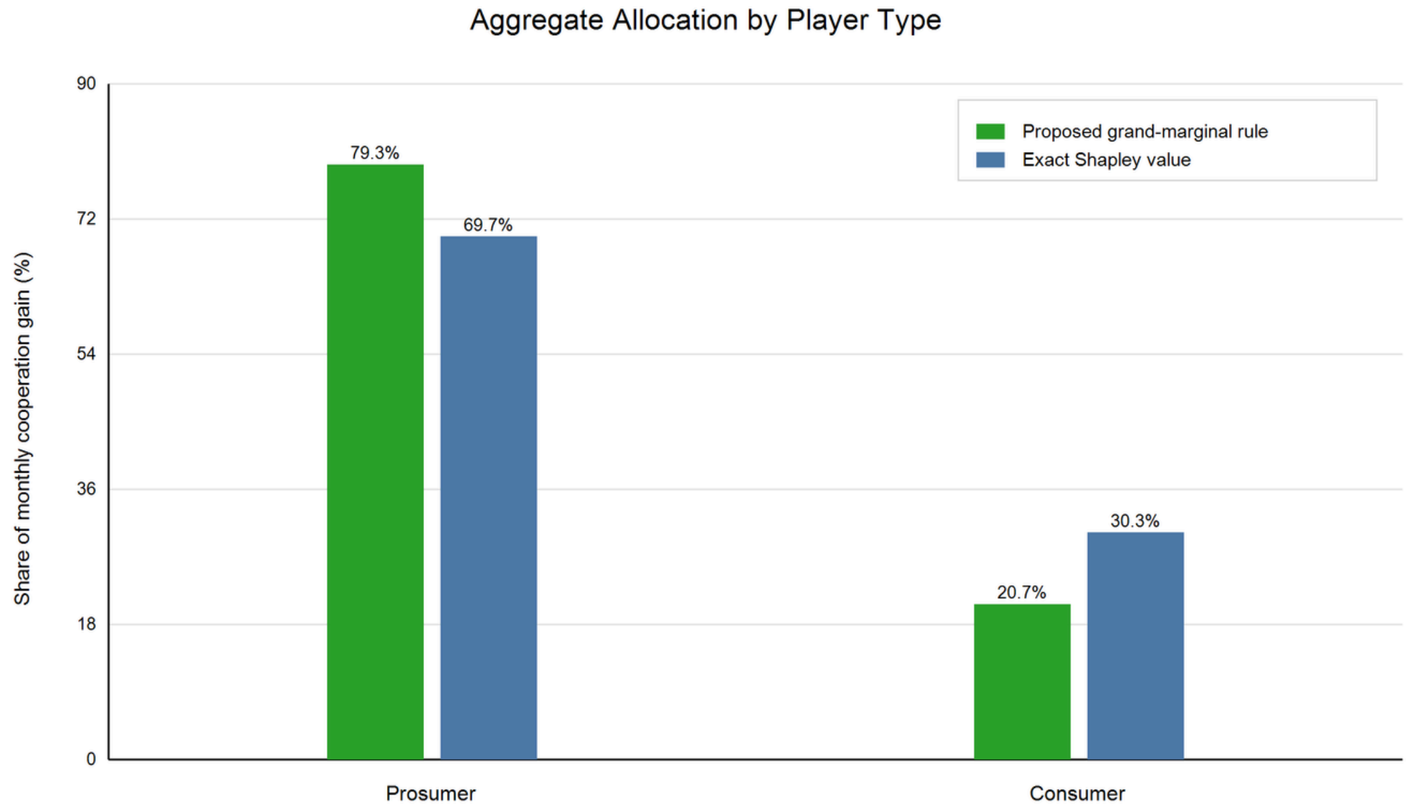


Aggregate effect by player type

Prosumer–consumer view

- Proposed rule allocates 79.3% of the monthly gain to prosumers.
- Exact Shapley allocates 69.7% to prosumers.
- Consumers receive the remaining share: 20.7% vs. 30.3%.

Interpretation: The proposed rule gives a larger share to prosumers because it focuses on their marginal role in the grand coalition, while Shapley averages contributions across all coalition contexts.



Subcoalition Formation

Why subcoalitions?

- Grand coalition: optimal benchmark, but computationally and organizationally heavy.
- It requires a large daily MILP, broad information exchange, and centralized coordination.
- Objective: partition N users into K balanced groups that recover most of the grand-coalition benefit.
- Direct exhaustive grouping does not scale, so the search is guided by representative days, pairwise scores, and local search.

Formation pipeline

1

Representative days

Approximate the year

2

Pairwise scores

Estimate pairwise compatibility

3

Lexicographic grouping

Maximize weakest group score

4

Annual evaluation

Compare against grand coalition

Representative Day Selection

Goal: approximate annual behaviour while keeping pairwise-score computation tractable.

1 **Valid common days**

343 clean days with complete 96-period profiles.

2 **Daily net-load vectors**

Each day is represented by its aggregate net-load pattern.

3 **Extreme days**

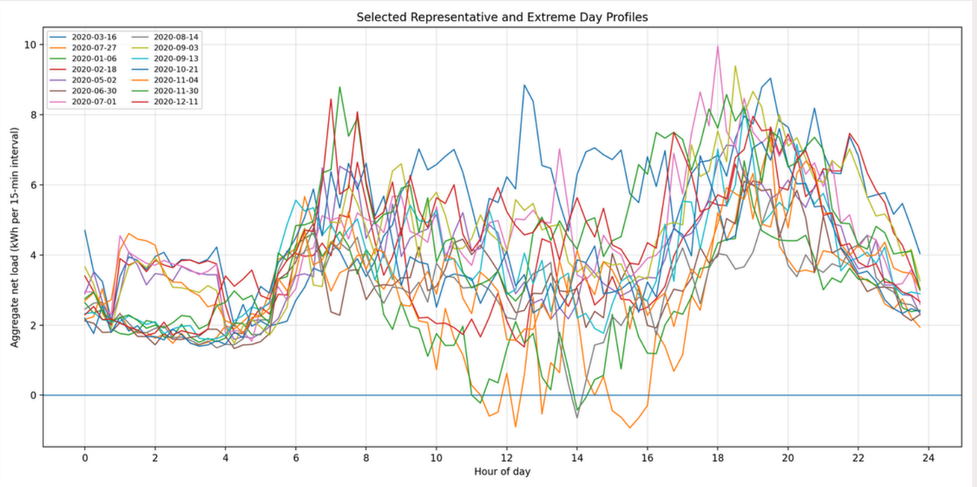
High-import and high-surplus days are included explicitly, with weight 1 each.

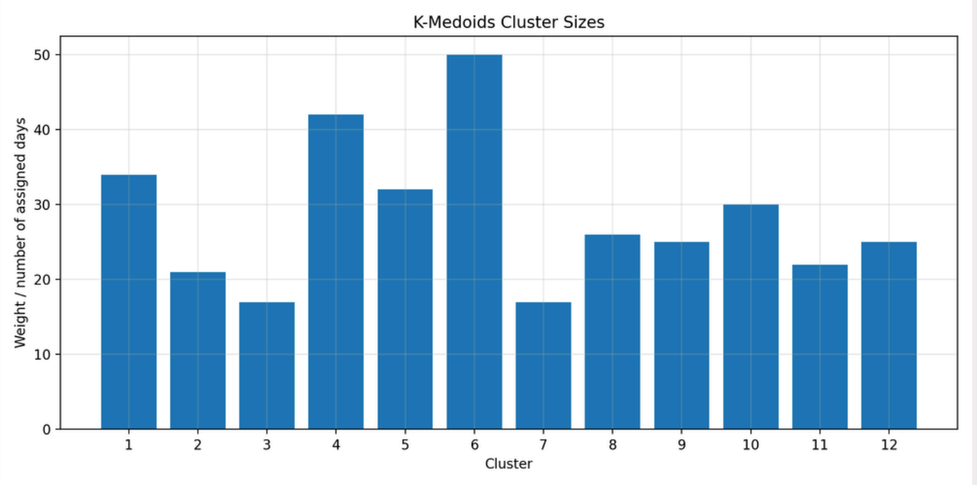
4 **k-medoids typical days**

After removing the two extremes, 12 actual medoid days are selected.

5 **Medoid weights**

Each medoid represents all days assigned to its cluster: $w_d = |\{d' : d' \text{ is assigned to medoid } d\}|$.





Pairwise Coordination Scores

Pairwise score

For each pair (i,j), compute their 2-user coordination gain on each representative day.

Weighted pairwise score

$$\sigma_{ij} = \sum_{d \in \mathcal{D}_{\text{rep}}} w_d \cdot g_{ij}(d)$$

medoids: w_d = |{d' : d' is assigned to medoid d}|
extreme days: w_d = 1

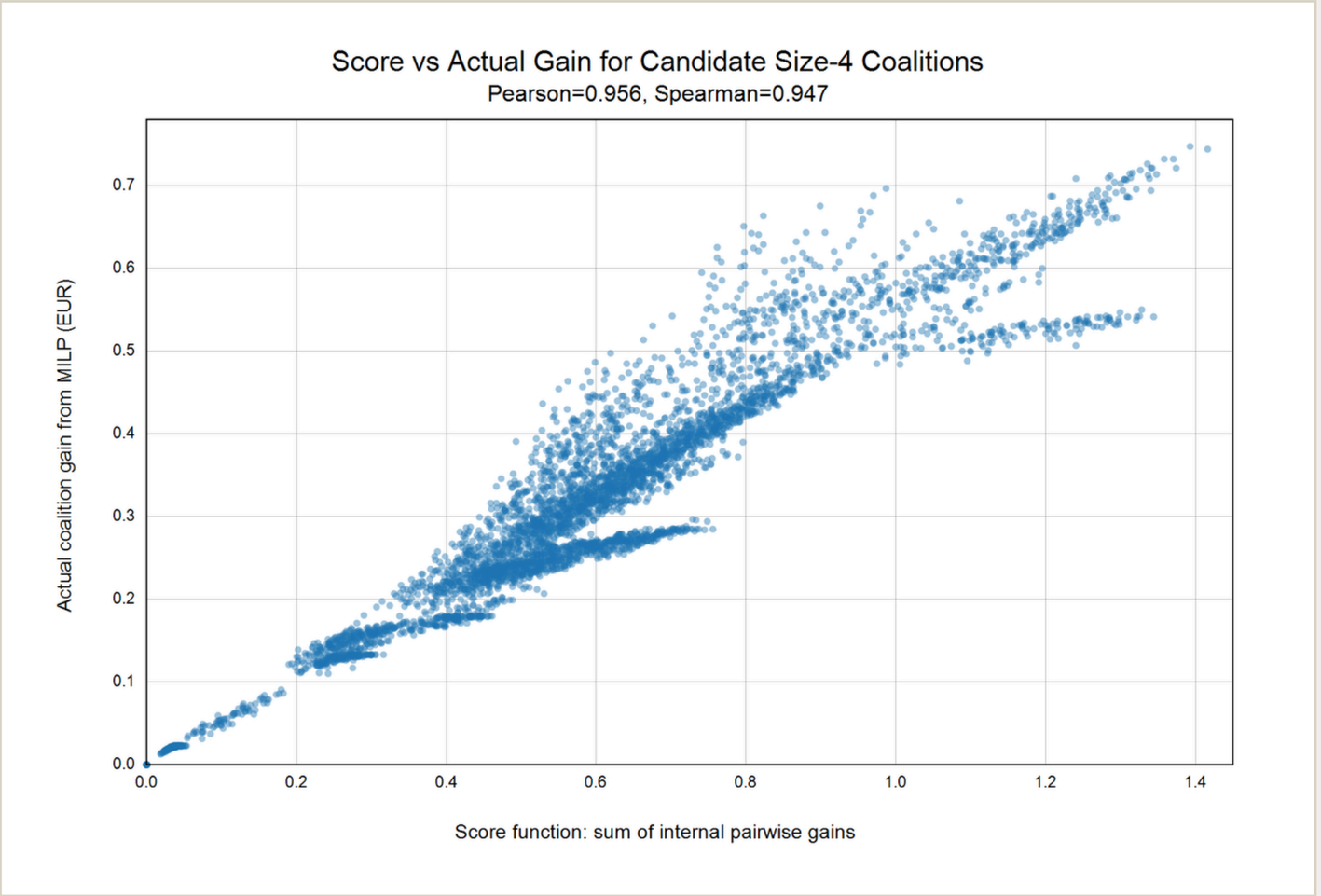
Coalition score

$$\text{Score}(S) = \sum_{\substack{i<j \\ i,j \in S}} \sigma_{ij}$$

Validation

For candidate size-4 coalitions, the score is strongly aligned with the actual MILP coalition gain:

Pearson = 0.956, Spearman = 0.947



Lexicographic Fairness & Exact MILP

Why lexicographic fairness?

The objective is not only to maximize total score. A partition with a high total score may still contain one weak subcoalition.

Lexicographic objective

1. Maximize the weakest group score:

$$\max \min_k S_k$$

2. Among ties, maximize total score:

$$\max \sum_k S_k$$

Fair: no group is left behind. **Efficient:** remaining freedom improves the total.

Exact MILP grouping

Variables

$x_{ik} = 1$ if user i is assigned to group k
$y_{ijk} = 1$ if users i and j are both in group k

Group score

$$S_k = \sum_{i<j} \sigma_{ij} y_{ijk}$$

Constraints

each user assigned once · equal group size · y_{ijk} linearization

Two-stage solve

Stage 1: maximize z, with $z \le S_k$ for every group k
Stage 2: fix $z = z^*$ and maximize $\Sigma_k S_k$

Scalable Local Search

Why not the exact MILP?

The number of equal-size partitions explodes combinatorially:

$$\frac{n!}{(m!)^K K!}$$

20 players, 5 groups of 4

$\approx 3.05 \cdot 10^{11}$ partitions

30 players, 5 groups of 6

$\approx 1.43 \cdot 10^{19}$ partitions

Exact and optimal, but impractical as the number of players grows.

Multi-start local search

1. Start from many random feasible partitions.
2. Improve each one by swapping players between groups.
3. Accept a swap only if it improves the lexicographic objective: first min group score, then total score.
4. Repeat until no improving swap exists; thousands of restarts avoid poor local optima.

Same pairwise-score objective as the MILP — at a fraction of the cost.

30-player run

230.7 s

5000 starts · ≤150 passes · best found after 2773 starts

min 139.19 · max 140.51

σ = 0.48 → highly balanced groups

50-player runs

1222 – 1508 s

far cheaper than the exact MILP on the same instances

Experiment: Setup & Annual Results

Data & model

20 measured Irish household profiles (players 1–20) + synthetic LoadProfileGenerator profiles (players 21+).

15-minute resolution · 343 valid days of 2020 · same battery, tariff & cooperative-scheduling model as the grand coalition.

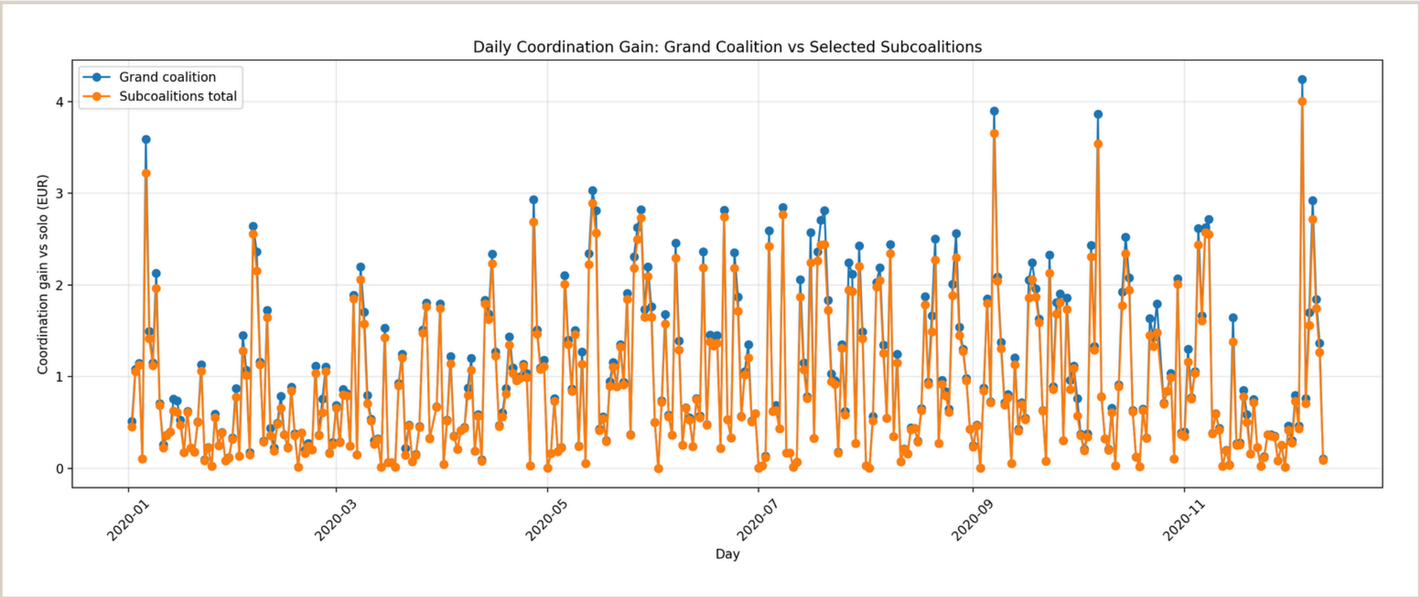
Three formation methods

1. Exact lexicographic MILP — tractable small cases
2. Pairwise-score local search — scalable 30- & 50-player cases
3. Balanced random benchmark — prosumers spread evenly, consumers random

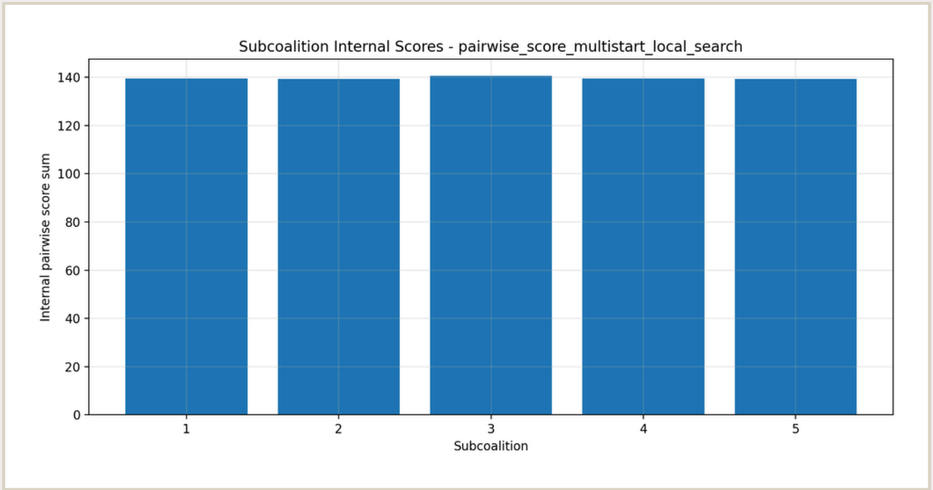
Case	Groups	Method	Grand gain (EUR)	Subcoal. gain (EUR)	Recovered (%)
16 players	4 × 4	Exact MILP	221.81	203.15	**91.59**
20 players	5 × 4	Exact MILP	297.47	259.84	**87.35**
30 players	5 × 6	Local search	331.81	312.72	**94.25**
50 players, 10P/40C	5 × 10	Local search	355.21	350.65	**98.72**
50 players, 25P/25C	5 × 10	Local search	601.59	570.32	**94.80**
50 players, 25P/25C	10 × 5	Local search	601.59	538.12	**89.45**

Subcoalitions recover 87–99% of the grand-coalition coordination gain across all cases.

Experiment: Results & Random Benchmark



Daily gain of the 30-player local-search subcoalitions tracks the grand coalition closely over the whole year.



Internal scores: min 139.19, max 140.51, σ = 0.48 — a highly balanced partition.

vs balanced random benchmark (prosumers evenly split — a strong baseline, not pure chance)

91.6% vs 83.7%

16 players — score-based clearly beats random: in small communities, who is grouped with whom matters.

94–99% ≈ random

Large, balanced groups (30 & 50 players, 5 groups) — prosumer/consumer balance already captures most of the value.

89.5% vs 85.4%

10 groups of 5 — score-based beats even the best random draw (85.9%). Details on the next slide.

50 Players (25P/25C): Group Size Matters

Same population, same grand-coalition gain (601.59 EUR) — only the partition changes.

5 groups of 10

94.80%

of the grand-coalition gain recovered

Subcoalition gain: **570.32 EUR**

Balanced random: mean 94.06% · best draw 94.80%

Score-based ≈ random — balance already captures most of the value.

10 groups of 5

89.45%

of the grand-coalition gain recovered

Subcoalition gain: **538.12 EUR**

Balanced random: mean 85.41% · best draw 85.86%

Score-based wins by +4.0 pp — above even the best random draw.

Why? Smaller groups restrict internal energy exchange, so less of the grand coalition’s flexibility survives (89.45% vs 94.80%) — and with fewer members per group, individual matching matters more, which is exactly where pairwise-score optimization pays off.

Stability under Profit Sharing: Accepted Migrations

The question

After the daily gains are allocated with the positive leave-one-out rule, does the weakest member of each subcoalition have a credible incentive to leave?

Candidate mover: the weakest member

$$w_k = \arg\min_{i \in S_k} B_i(S_k)$$

A move is a blocking accepted migration if:

1. the moving player strictly improves:

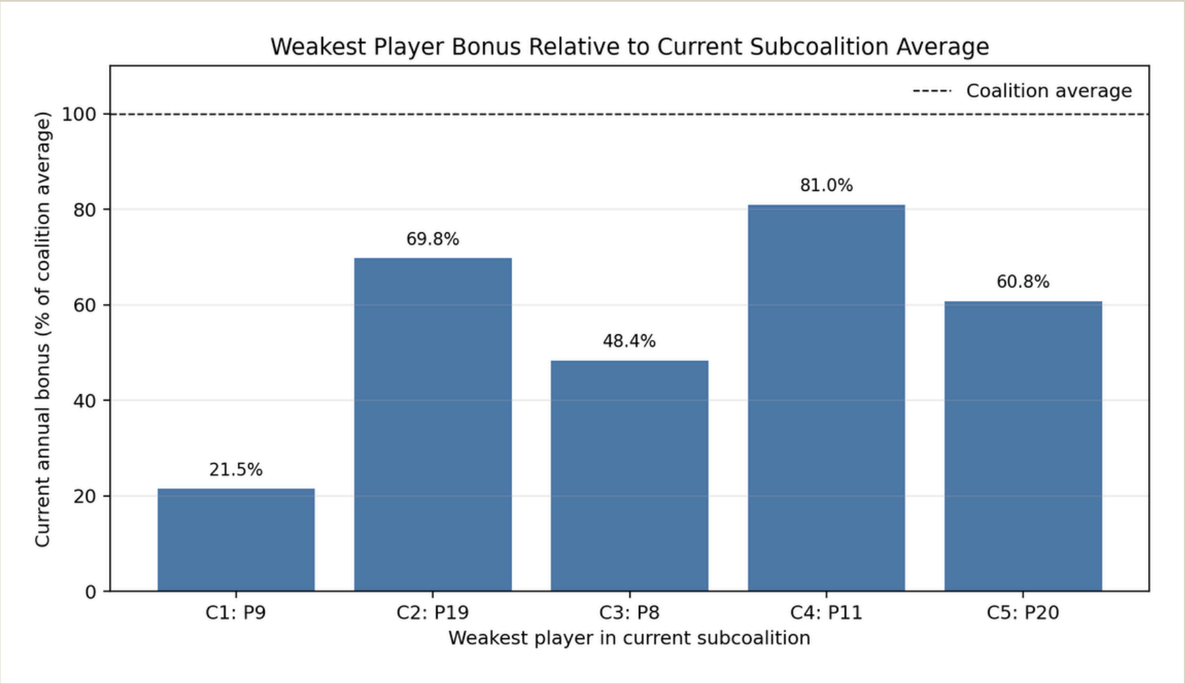
$$B_{w_k}(S'_\ell) - B_{w_k}(S_k) > 0$$

2. every current member of the target coalition weakly improves:

$$B_j(S'_\ell) - B_j(S_\ell) \geq 0 \quad \forall j \in S_\ell$$

No unilateral entry: the mover needs the target's consent. The source coalition has no veto.

Setup: 20 players · 5 subcoalitions from the pairwise-score formation · 343 days of 2020 · 5 movers × 4 targets = 20 candidate migrations



Player 9 (C1) earns only 21.5% of its group's average bonus — the most natural candidate to seek another group.

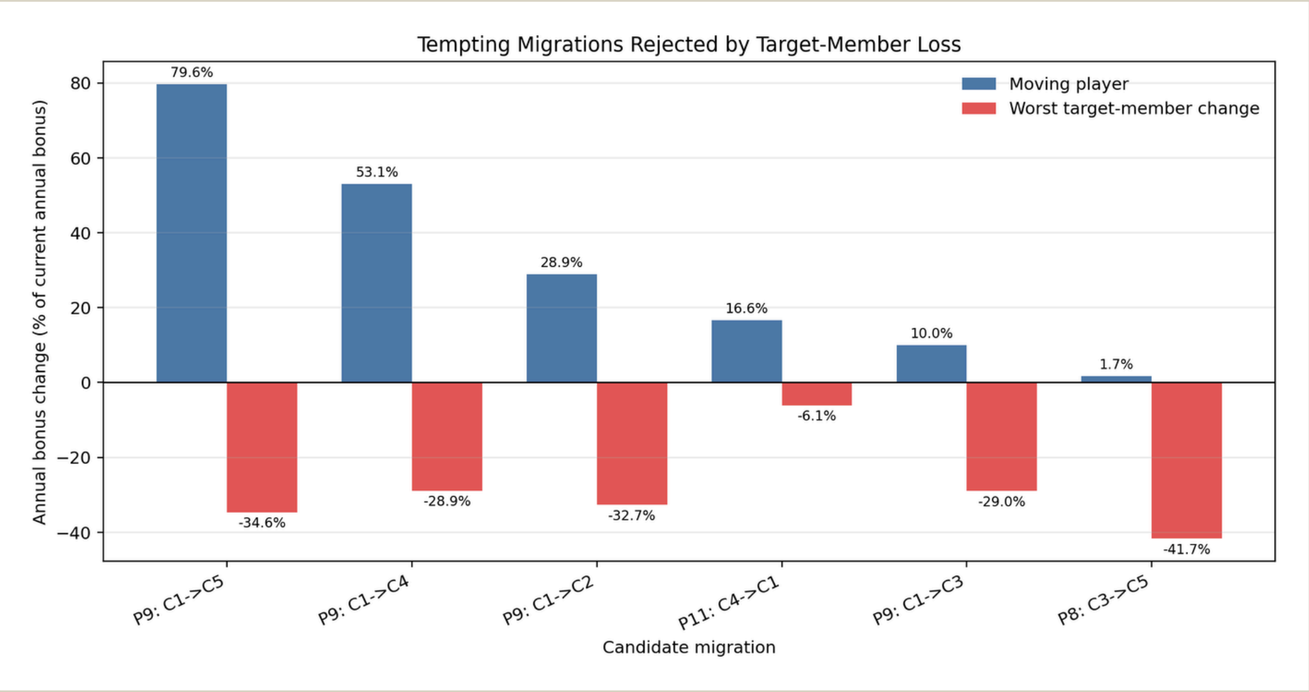
Stability Results: No Accepted Migration

20
candidate migrations checked

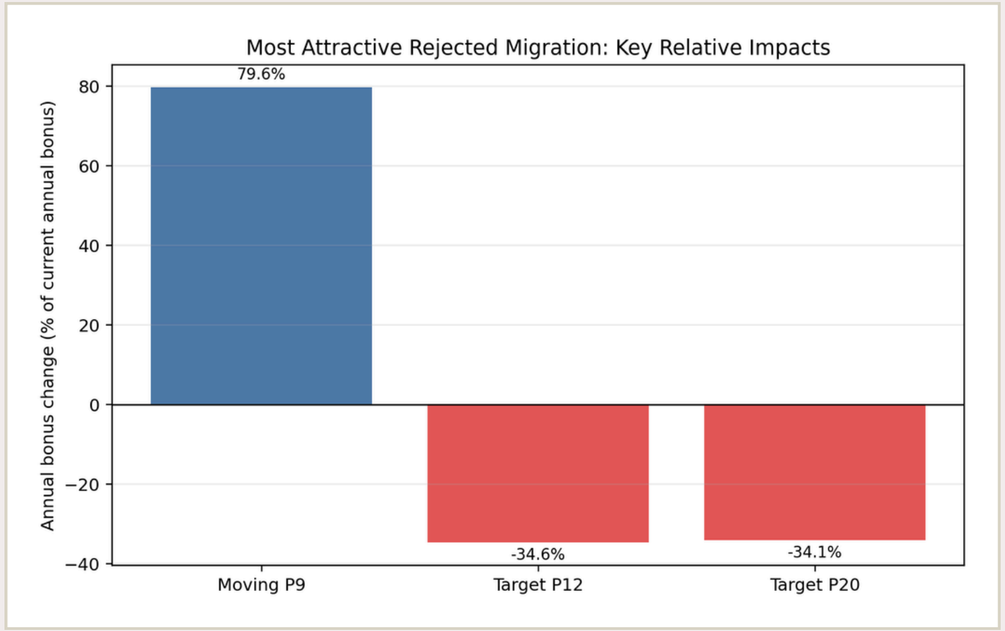
6
would benefit the mover

+79.6%
largest tempting gain (P9: C1→C5)

0
blocking accepted migrations



Every tempting move is rejected: at least one current target member would lose annual bonus.



Best rejected move — P9: C1→C5: mover +79.6%, but P12 −34.6% and P20 −34.1% → the target refuses.

Demand Response: Targeted Load Shifting

Idea: shift a small amount of controllable demand away from the most expensive hour and recover it later — load shifting, not shedding.

1. **Activation window**
   The 1-hour window (4 × 15-min) with the largest aggregate import cost of the day.

2. **Cap the heavy users**
   Each user is capped at 80% of the mean activation-window consumption; only the excess is shifted, proportionally.

3. **Battery-aware recovery**
   The shifted energy returns in a 2-hour window chosen by post-battery marginal cost — not just the cheapest price, since the battery may have already captured that opportunity.

4. **Re-solve & measure**
   Battery scheduling is re-solved on the modified profiles; the DR benefit is the cost difference vs. the baseline schedule.

Setup

20 users · 343 days · 15-min

Cap: 80% of mean activation load
Activation: 1 h · Recovery: 2 h
Recovery rule: battery-aware

Central ideal battery: 0–80 kWh

Demand Response: Modest at Household Scale

No battery

−2.0%

2 834 kWh shifted · ~8.7 users/day · 0.048 €/kWh

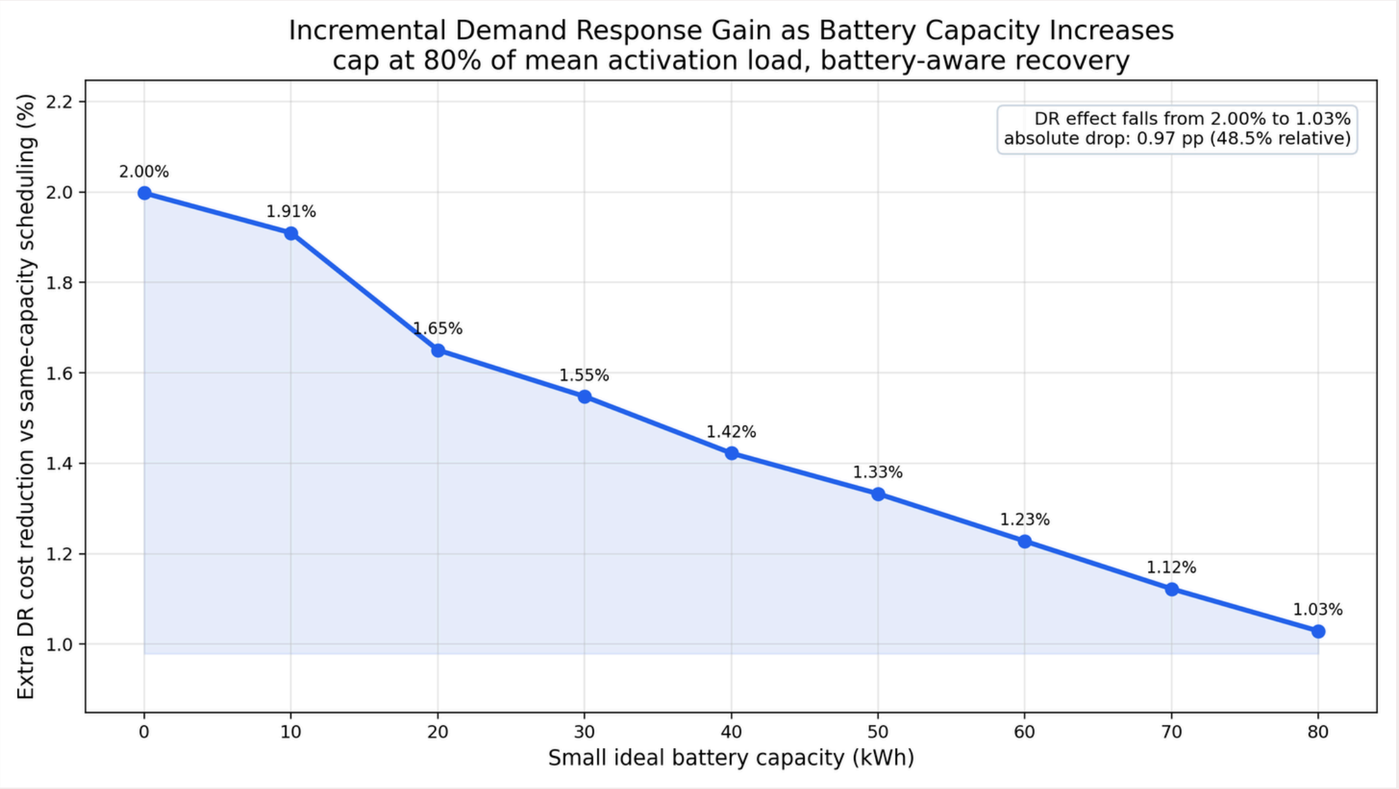
With an 80 kWh battery

−1.0%

same shifted energy, less than half the value (0.021 €/kWh)

Why it shrinks: battery and DR compete for the same arbitrage — as storage grows, most of the opportunity is already captured by the schedule.

80 kWh + DR reaches **70.4%** of the 185 kWh reference gain



Perspective: modest for households — but at operator scale (negative prices, MWh volumes, imbalance exposure) demand response becomes dispatchable flexibility with a different value proposition.

Conclusions

1 Cooperation has measurable value

12.2% lower annual cost vs. baseline: 8.1% from individual batteries **+ 4.1 pp added by coordination** — two distinct, additive value sources.

2 The surplus can be shared fairly — at scale

The leave-one-out rule needs **2|N| + 1 problems** (93% fewer solves than Shapley in the 8-player test), stays close to Shapley, and satisfies the properties that matter in practice.

3 Subcoalitions preserve the value

87–99% of the grand-coalition gain recovered. Prosumer/consumer balance is first-order; pairwise scores pay off when groups are small (89.5% vs 85.4% random).

4 The groups are stable after allocation

0 accepted migrations over the full year — 30% of moves would tempt the mover, yet every one is refused by the receiving group.

5 Demand response: complement, not substitute

2.0% savings without battery, shrinking to **1.0%** with 80 kWh — battery and DR compete for the same arbitrage opportunity.

Future work

- Stochastic / robust scheduling under price & forecast uncertainty
- Distribution-grid constraints — voltage, congestion, losses
- Stronger stability notions — all unilateral moves, swaps, collective deviations
- Demand response & flexible loads inside the cooperative framework
- More datasets, tariff schemes & community compositions

Takeaway: *energy communities are coordinated systems, not collections of independent assets — one unified, scalable, interpretable framework links operation, allocation, formation and incentives.*

Dataset & Experimental Setup

15-min
resolution, year 2020

96
time steps per day

343
clean common days
(02 Jan – 11 Dec · 367 − 24 removed)

Data

Measured Irish energy-community households (StoreNet, Dingle) *(Trivedi et al., 2024)*, plus synthetic LoadProfileGenerator profiles for the larger cases.
Per user and 15-min step: production · consumption · net load (= consumption − production)

Prices

Irish day-ahead market prices per 15-min interval.
Buy and sell prices are derived from the day-ahead price, with **buy > sell** at all times.

Battery — per player

10 kWh · 3.3 kW
capacity · max charge / discharge

95% round-trip efficiency
85% DoD → 8.5 kWh usable

SoC_0 = 50% (4.25 kWh)
terminal SoC = initial, each day

Grid import / export ≤ 20 kW

Thank You